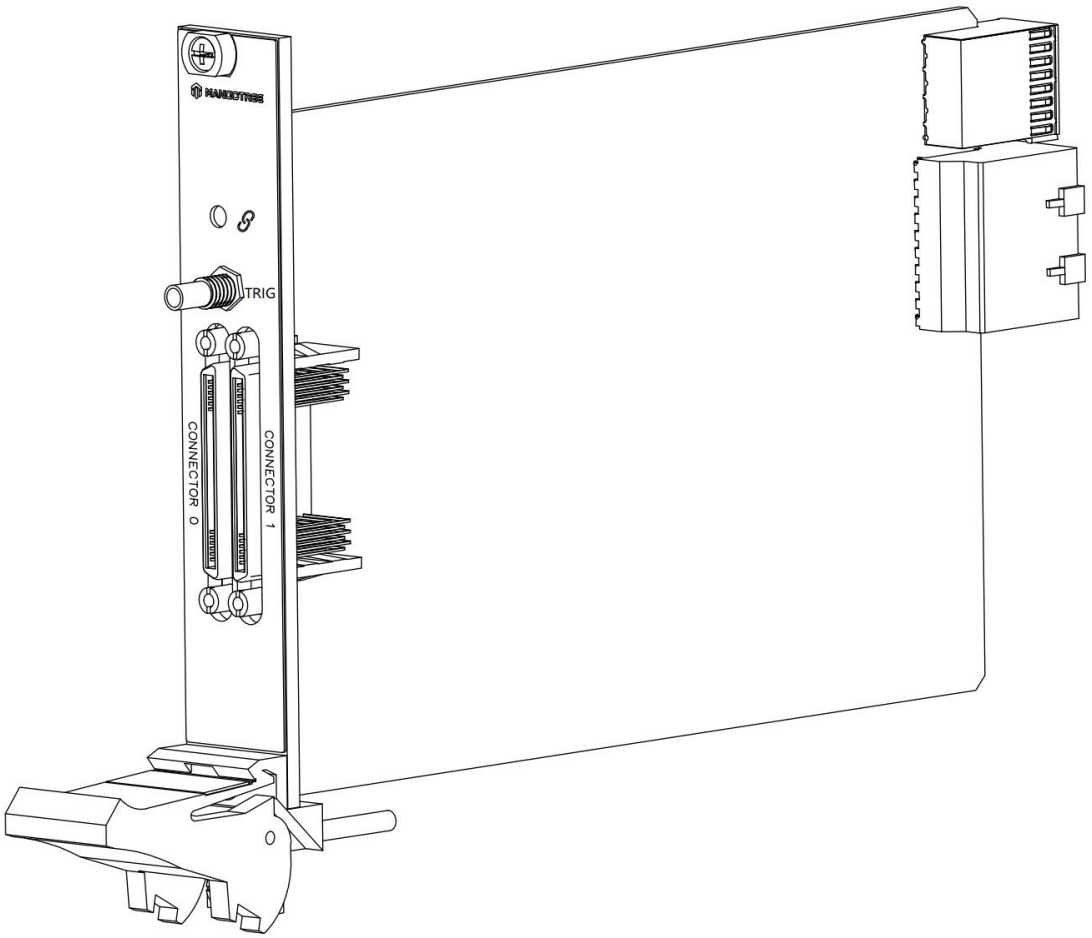
DataSheet

# MT-X532

This document contains the specifications for MT-X532.Specifications are typical at 25°C unless otherwise noted.

**Caution** Using the MT-X532 in a manner not described in this document may impair the protection the MT-X532 provides.




MANGOTREE

| | |
|---|---|
| Number of Channels | 4 |
| Counter measurements | Edge counting,pulse,pulse width,frequency |
| Maximum input frequency | 2MHz |

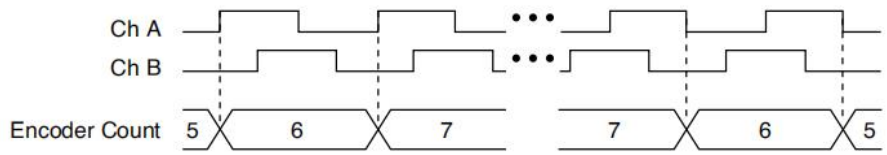
# PWM

| | |
|---|---|
| Voltage Level | 5V |
| Number of Channels | 4 |
| Maximum output frequency | 500kHz |

# Encoder

| | |
|---|---|
| Voltage Level | 5V |
| Number of Channels | 2 (each with A,B,Z) |
| Position measurements | Quadrature encoding |

**Figure 1.** Quadrature encoding




# Digital Trigger

| | |
|---|---|
| Voltage Level | 5V |
| Number of Channels | 2 |

# Digital Waveform Input/Output

| | |
|---|---|
| Voltage Level | 5V |
| Number of Channel | 2 Digital Waveform In, 2 Digital Waveform Out |
| Maximum update rate | 5MHz |

# Digital Input/Output

| | |
|---|---|
| Voltage Level | 5V |
| Number of Channel | 6 Digital Input,6 Digital Output |
| Maximum update rate | 500Hz |

# External Digital Triggers

| | |
|---|---|
| Number of triggers | 12 |
| Source | Digital Triggers,TRIG,PXI_TRIG<0..7>, PXI_STAR |

# Maximum Power Requirements

Power requirements are dependent on the digital output loads and configuration of the LabVIEW FPGA VI used in your application.

| | |
|---|---|
| +3.3V | 1.23 A |
| +12 V | 0.57 A |

# Physical Characteristics

| | |
|---|---|
| Weight | 200g |
| Dimensions | See end of this document |

# Safety Voltages

This product is designed to meet the requirements of the following electrical equipment safety standards for measurement, control, and laboratory use:

- IEC 61010-1, EN 61010-1
- UL 61010-1, CSA C22.2 No. 61010-1

# CE Compliance CE

This product meets the essential requirements of applicable European Directives, as follows:

- 2014/35/EU; Low-Voltage Directive (safety)
- 2014/30/EU; Electromagnetic Compatibility Directive (EMC)
- 2014/34/EU; Potentially Explosive Atmospheres (ATEX)

# Shock and Vibration

To meet these specifications, you must panel mount the system.

| | |
|---|---|
| Operating vibration | |
| Random (IEC 60068-2-64) | 5 $g_{rms}$, 10 Hz to 500 Hz |
| Sinusoidal (IEC 60068-2-6) | 5 g, 10 Hz to 500 Hz |
| Operating shock (IEC 60068-2-27) | 30 g, 11 ms half sine; 50 g, 3 ms half sine;<br>18 shocks at 6 orientations |

# Environmental

Refer to the manual for the chassis you are using for more information about meeting these specifications.

| | |
|---|---|
| Operating temperature | -40 °C to 70 °C<br>(IEC 60068-2-1, IEC 60068-2-2) |
| Storage temperature | -40 °C to 85 °C<br>(IEC 60068-2-1, IEC 60068-2-2) |
| Ingress protection | IP40 |
| Operating humidity (IEC 60068-2-78) | 10% RH to 90% RH, noncondensing Storage |
| humidity (IEC 60068-2-78) | 5% RH to 95% RH, noncondensing Pollution |
| Degree | 2 |
| Maximum altitude | 4,000 m |

Indoor use only.

# DAQMode Support

通用模式（DAQMode=0）包括：AI、AO、Counter、PWM、Encoder、Digital Input、Digital Output、Digital Waveform Input、Digital Waveform Output、Temperature。

通用模式适用所有的MT-DAQ设备，该模式下用户可以开发使用设备数据手册中说明的全部功能，不同设备支持的功能不同。本设备通用模式支持的功能如下表所示，对应功能为绿色则支持，红色则不支持。

本设备通用模式（DAQMode=0）支持的功能：

| | |
|---|---|
| 🔴AI | 🟢AO |
| 🟢Counter | 🟢PWM |
| 🟢Encoder | 🔴Temperature |
| 🟢Digital Input | 🟢Digital Output |
| 🟢Digital Waveform Input | 🟢Digital Waveform Output |

特殊模式包括：AI/AO同步（AIO Sync）、编码器触发AI（Encoder Trigger AI）、AI触发编码器（AI Trigger Encoder）、DI触发AI/AO同步（DI Trigger AIO）、DI触发AI（DI Trigger AI）、DI触发AO（DI Trigger AO）。

如果需要使用特殊模式下的模式，用户需要确认购买设备是否支持该模式，本设备支持的特殊模式如下表所示，对应模式为绿色则支持，红色则不支持；Customer是定制DAQ模式，正常用户无法使用。如果模式配置错误，会导致设备无法正常运行。

本设备支持的特殊模式：

| | |
|---|---|
| 🔴AO-Sync-AI (DAQMode=1) | 🔴AI-Trigger-Encoder (DAQMode=2) |
| 🔴Encoder-Trigger-AI (DAQMode=3) | 🔴DI-Trigger-AI-Sync-AO (DAQMode=4) |
| 🔴DI-Trigger-AI (DAQMode=5) | 🟢DI-Trigger-AO (DAQMode=6) |

# Config文本

MT-DAQ设备的开发和使用依赖于Config配置文本，只有正确配置该文本，才能保证设备的正常运行。不同型号的设备或板卡对应的配置参数是不同的。Python、LabVIEW和C#三种编程语言的Config配置文本完全相同。

通用Config配置文本通过MT-Master软件主页导出获得，用户可以根据实际设备或板卡的参数对配置文本进行修改配置，或者按照文本默认参数配置运行。

Config配置文本中的各项参数含义及其具体配置可以参考MT-DAQ上手指南，指南链接附于下文Support板块。

使用MT产品过程中如有任何疑问，可以通过访问官网：http://www.mangotree.cn联系专业客服咨询。

MangoTree官网

# Support

MT-Master上手指南：

https://server.mangotree.cn:9900/WebFile/Downloads/上手指南/MT-Master/

Master上手指南

MT-Master视频教程：

https://server.mangotree.cn:9900/WebFile/Downloads/视频教程/MT-Master/

Master视频教程

MT-DAQ上手指南：

https://server.mangotree.cn:9900/WebFile/Downloads/上手指南/MT-DAQ/

DAQ上手指南

MT-DAQ视频教程：

https://server.mangotree.cn:9900/WebFile/Downloads/视频教程/MT-DAQ/

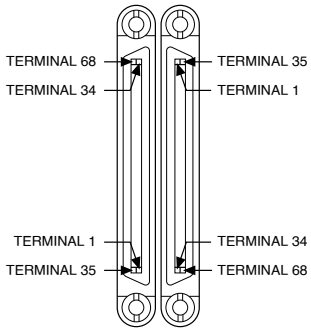
DAQ视频教程

# MT-X532 Pinout

CONNECTOR 0

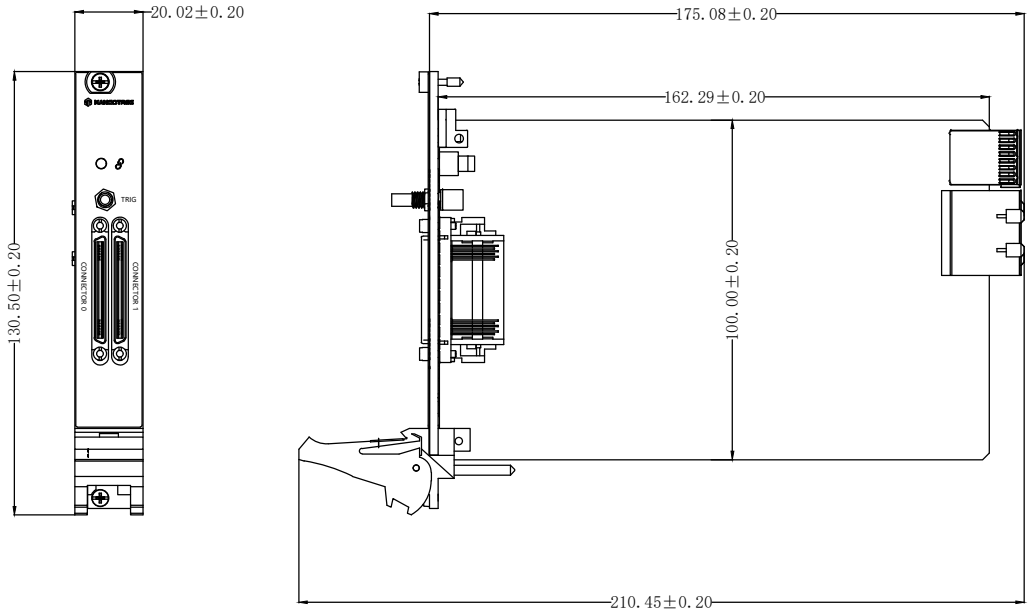
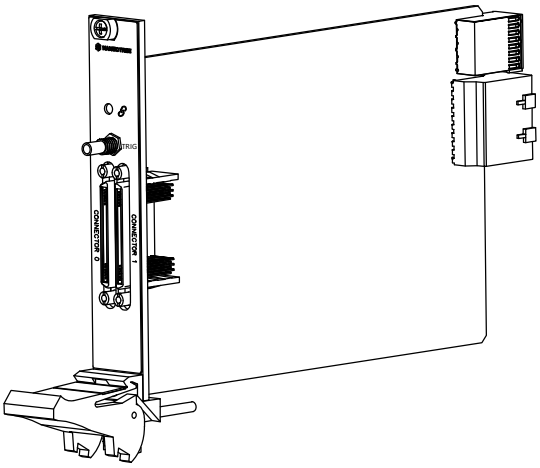
| | | | |
|---|---|---|---|
| AO 0 | 68 | 34 | NC |
| A GND | 67 | 33 | A GND |
| AO 1 | 66 | 32 | NC |
| AO 2 | 65 | 31 | NC |
| A GND | 64 | 30 | A GND |
| AO 3 | 63 | 29 | NC |
| AO 4 | 62 | 28 | NC |
| A GND | 61 | 27 | A GND |
| AO 5 | 60 | 26 | NC |
| AO 6 | 59 | 25 | NC |
| A GND | 58 | 24 | A GND |
| AO 7 | 57 | 23 | NC |
| A GND | 56 | 22 | A GND |
| AO 8 | 55 | 21 | NC |
| AO 9 | 54 | 20 | NC |
| AO 10 | 53 | 19 | NC |
| AO 11 | 52 | 18 | NC |
| AO 12 | 51 | 17 | NC |
| AO 13 | 50 | 16 | NC |
| AO 14 | 49 | 15 | NC |
| AO 15 | 48 | 14 | NC |
| Digital Trigger 0 | 47 | 13 | Digital Waveform Out 0 |
| Digital Waveform In 0 | 46 | 12 | Digital Output 2 |
| Digital Output 1 | 45 | 11 | Digital Output 0 |
| Digital Input 2 | 44 | 10 | Digital Input 1 |
| Digital Input 0 | 43 | 9 | D GND |
| Counter 1 | 42 | 8 | D GND |
| Counter 0 | 41 | 7 | D GND |
| PWM 1 | 40 | 6 | D GND |
| PWM 0 | 39 | 5 | D GND |
| Encoder0.Z | 38 | 4 | D GND |
| Encoder0.B | 37 | 3 | D GND |
| Encoder0.A | 36 | 2 | D GND |
| +5 V | 35 | 1 | +5 V |

NC = No Connect




CONNECTOR 1

| | | | |
|---|---|---|---|
| +5 V | 1 | 35 | +5 V |
| D GND | 2 | 36 | Encoder1.A |
| D GND | 3 | 37 | Encoder1.B |
| D GND | 4 | 38 | Encoder1.Z |
| D GND | 5 | 39 | PWM 2 |
| D GND | 6 | 40 | PWM 3 |
| D GND | 7 | 41 | Counter 2 |
| D GND | 8 | 42 | Counter 3 |
| D GND | 9 | 43 | Digital Input 3 |
| Digital Input 4 | 10 | 44 | Digital Input 5 |
| Digital Output 3 | 11 | 45 | Digital Output 4 |
| Digital Output 5 | 12 | 46 | Digital Waveform In 1 |
| Digital Waveform Out 1 | 13 | 47 | Digital Trigger 1 |
| A GND | 14 | 48 | AO 31 |
| A GND | 15 | 49 | AO 30 |
| A GND | 16 | 50 | AO 29 |
| A GND | 17 | 51 | AO 28 |
| A GND | 18 | 52 | AO 27 |
| A GND | 19 | 53 | AO 26 |
| A GND | 20 | 54 | AO 25 |
| A GND | 21 | 55 | AO 24 |
| A GND | 22 | 56 | A GND |
| NC | 23 | 57 | AO 23 |
| A GND | 24 | 58 | A GND |
| NC | 25 | 59 | AO 22 |
| NC | 26 | 60 | AO 21 |
| A GND | 27 | 61 | A GND |
| NC | 28 | 62 | AO 20 |
| NC | 29 | 63 | AO 19 |
| A GND | 30 | 64 | A GND |
| NC | 31 | 65 | AO 18 |
| NC | 32 | 66 | AO 17 |
| A GND | 33 | 67 | A GND |
| NC | 34 | 68 | AO 16 |

NC = No Connect

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